

Istiak Hassan Emon

Engineering Lead · Platform & Distributed Systems · Rust, Go, Kubernetes
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Summary

Engineering lead who designs and runs platforms that must work at scale: Rust and Go systems, Kubernetes, GitOps, on-prem fleets. I lead four product lines and shipped an imaging backbone (70k studies/month), a fleet control plane (86 servers) and a delivery pipeline with ~1,500 releases in 18 months.

Client names and references available on request (under NDA). Metrics come from production telemetry.

Experience

TRODAD International Ltd — Dhaka, Bangladesh
Development & Technology Team Lead — February 2025 – Present
Health-technology group: engineering lead across a national radiology platform, a multi-site ERP, a lab-integration product and a consumer health platform.

- Lead engineering for 4 product lines: built and run the hiring loop, mentor engineers into owners of production services, and set the architecture and code-review standards every team ships on.
- Defined the release process (trunk + release PRs, release-please, goreleaser, Tauri auto-update) that shipped ~1,500 tagged releases in 18 months across 18 core repos.
- Built a medical-imaging (DICOM) server in Rust from scratch for a national radiology platform: volume grew 7× in 6 months to 70,651 studies/month from 375 active sites at p95 latency 234 ms, with 6.3M images (4.5 TB) stored and 95 releases (DIMSE + DICOMweb, S3/MinIO, Redis circuit breaker, Prometheus, audit logs); plus a Rust/Tauri site router (35 releases) with a persistent queue and 5 codecs incl. JPEG-XL.
- Resolved a 2-hour radiology worklist outage (workflows/hour 827 to 54, lag up to 1.9 h) by tracing a four-link failure chain (unindexed cleanup scan, lock convoy, misleading healthcheck, restart loop) and shipping an index, a bounded resumable task, a healthcheck fix, a rolling-window restart breaker and new lag and lock alerts.
- Designed and led a Go fleet control plane for an on-prem ERP live at 88 sites on 86 servers (85 on a WireGuard mesh, 600k+ patient records): automatic pull updates keep 83 of 88 sites on the current release, with an audited remote terminal instead of shared SSH.
- Cut manual lab-result entry with a Rust/Tauri bridge whose instrument parsers compile to WASM (ASTM/HL7, 17 bidirectional order modules, QC, licensing; 90 releases): live at 8 labs, ~23 analyzer families supported, 25,000+ analyzer results captured since first production in June 2026.
- Made rollout repeatable with a one-binary Go deployer (29 releases) that brings up the full stack (Traefik, Django/Celery, PACS, Postgres, Redis, RabbitMQ, MinIO) with automatic DNS, and a production k3s + ArgoCD GitOps cluster for one product line (Longhorn, CloudNativePG, MinIO, kube-prometheus; 35 app manifests). Designed a 5-zone PHI de-identification pipeline (Rust/Axum + Go) so imaging data can feed AI training safely.

Nexis Ltd — Dhaka, Bangladesh
Founder — September 2021 – Present

- Founded a software company (full-time until Feb 2025, part-time since then alongside TRODAD) delivering multi-tenant SaaS (HRM, school management, digital menus, e-commerce, SSO), owning architecture, delivery and client relationships.
- Built the infrastructure-as-code foundation reused by every product (889 commits): Terraform across Oracle Cloud, Proxmox and AWS EKS, with Packer and Ansible.

- Launched Lepti (2026), a VPN/proxy service, from zero: Go API (billing, seats, passkeys/TOTP), Go node agent (xray/hysteria, nftables) and a Rust desktop client.
- Built a Go logistics backend and published open-source npm packages (country, date2word).

myBurgerLab Malaysia
Systems Support Assistant January 2020 – June 2021

ZE Enterprise Shah Alam, Malaysia
Web Developer Intern May 2019 – August 2019

Selected Work

Rust imaging pipeline at national scale
Rust PACS, site router and PHI de-identification hub moving 70k studies/month from 375 active sites (p95 234 ms, 6.3M images).
ihemon.me/work/rust-imaging-pipeline

Fleet control plane for 86 on-prem servers
Go hub, agents and WireGuard mesh for an ERP live at 88 sites on 86 servers; keeps 83 of 88 sites on the current release with audited remote access.
ihemon.me/work/fleet-control-plane

Lepti: a VPN product from zero
Go control API and node agents plus a Rust desktop client; billing, seats, passkeys/TOTP.
ihemon.me/work/lepti-vpn

Plugin-based Rust + WASM device integration engine
Rust/Tauri bridge with per-device WASM parsers over ASTM/HL7: live at 8 labs, ~23 analyzer families supported, 25,000+ analyzer results captured since June 2026; 90 releases.
ihemon.me/work/device-integration-engine

Skills

Languages
Rust, Go, TypeScript, Python, PHP, SQL, Shell/PowerShell

Backend & systems
Distributed services, DICOM (DIMSE/DICOMweb), HL7, ASTM, Axum, Gin, Django/DRF, Celery, RabbitMQ, Tauri, WebAssembly plug-ins, Next.js, Laravel

Platform & infrastructure
Kubernetes (k3s), ArgoCD/GitOps, Terraform, Packer, Ansible, Docker, Traefik, Longhorn, WireGuard, Prometheus/Grafana/Loki, GitHub Actions, release-please, goreleaser, On-prem fleet management, AWS EKS, Oracle Cloud, Proxmox

Data
PostgreSQL (CloudNativePG), Redis, S3/MinIO, Message queues, Data pipelines, Large schema migrations

Security
PHI de-identification, Passkeys/WebAuthn, TOTP, Audited remote access, nftables, Secure release & auto-update, Security background: offensive testing and responsible disclosure

Education

SEGi University BIT
Computer Networking 2021

SEGi College Diploma in IT
2019

Languages

Bengali	**English**	**Hindi**	**Bahasa Melayu**
Native	Fluent	Proficient	Conversational